

ANH-DUNG HUYNH

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Nha Trang, Vietnam (Relocating to Ho Chi Minh City)

SUMMARY

AI Engineer specializing in Agentic Orchestration (LangGraph, Spring Boot), advanced RAG (hierarchical chunking, document graphs), and production-grade asynchronous backends (Python, Java, RabbitMQ, Kafka, modern vector databases). Proven track record leading cross-functional teams and deploying enterprise conversational systems, real-time multimodal roleplay simulations, and high-precision legal NLP pipelines.

EDUCATION

UNIVERSITY OF INFORMATION TECHNOLOGY, VNU-HCM | *B.S. in Computer Science* Sept 2022 – Sept 2025

- 2nd Place - UIT Data Science Challenge 2024

EXPERIENCE

FPT SOFTWARE

Vietnam

AI Engineer

Sept 2025 – Present

- Led a 5-person AI development team, collaborating with PM, PO, and external vendors across iterative delivery cycles to deploy enterprise AI automation systems.
- Took end-to-end technical ownership of a production conversational workflow system automating internal workflows and structured ticket orchestration from architecture through cloud deployment.
- Architected a multi-layer query routing engine combining LLM reasoning, semantic vector similarity, and deterministic rule matching to bypass redundant generative model invocations and optimize downstream processing.
- Architected an asynchronous, message-driven backend with RabbitMQ to decouple external client adapters (e.g., MS Teams) from heavy inference workloads, eliminating synchronous HTTP connection timeouts through instant event acknowledgment and streamed ephemeral status updates.
- Solo-designed an agentic orchestration platform in Java (Spring Boot) featuring a declarative dynamic tool-dependency graph that calculates shortest execution paths based on parameter availability, context history, and vector retrieval (Qdrant).
- Enforced enterprise-grade reliability using Pydantic validation, structured exception hierarchies, distributed tracing, and centralized logging across Python and Java microservices.

HPT CORPORATION

Ho Chi Minh City, Vietnam

AI Engineer Intern

Jan 2025 – Jun 2025

- Spearheaded Legal NLP R&D, designing high-precision RAG pipelines and domain-adapted Named Entity Recognition (NER) models using PyTorch and Hugging Face transformers.
- Architected hierarchical document segmentation down to granular statutory units (articles, clauses, sections) using a custom regex engine to extract cross-reference citations and construct interconnected legal document graphs.

- Implemented Parent-Document Retrieval paired with hybrid search (lexical + dense) to overcome low semantic contrast and keyword confusion inherent in generalized embedding models on legal corpora.
- Addressed acute domain class imbalance via rule-based synonym substitution augmentation, enhancing token-level classification stability on niche legal entities.
- Benchmarked embedding architectures and vector representations, evaluating precision, recall, F1, and inference latency tradeoffs to guide production R&D strategy.

PROJECTS

DYNAMIC TOOL-ORCHESTRATION & GRAPH EXECUTION PLATFORM

2025 – Present

- Technologies: Java (Spring Boot), Python, Qdrant, Kafka, PostgreSQL, Redis
- Solo-architected an enterprise agent orchestration engine in Java (Spring Boot) with an admin portal enabling dynamic tool declaration, typed parameter schemas, and input-output aliasing.
- Modeled tool dependencies as dynamic directed graphs, computing optimal shortest execution paths (minimizing missing parameters) integrated with conversation history, user profiles, and vector retrieval (Qdrant).
- Built a high-throughput event-driven microservices architecture using Spring Boot, Kafka, and Qdrant, accompanied by a lightweight Python service for asynchronous knowledge ingestion.

PRODUCTION CONVERSATIONAL WORKFLOW & TICKET ORCHESTRATOR

2025 – Present

- Technologies: Python, LangChain, LangGraph, RabbitMQ, Milvus, Redis, Docker
- Led technical development for a production AI conversational engine automating enterprise workflows and internal support ticket routing.
- Designed a multi-tier routing pipeline combining deterministic matching, semantic vector search, and LLM reasoning to optimize response latency and token costs.
- Decoupled system components with a message-driven RabbitMQ bus, ensuring resilient, non-blocking execution under heavy concurrency.

AI INSURANCE AGENT MENTORING PLATFORM

2025 – Present

- Technologies: Python, LangGraph, FastAPI, D-ID, Qdrant, LLM-as-a-Judge, Docker
- Architected an interactive, real-time video simulation platform enabling insurance agents to practice client consultation through realistic conversational roleplay.
- Integrated dynamic conversational state graphs via LangGraph with streaming avatar and synthetic speech services (D-ID), managing bi-directional turn-taking and low-latency interaction.
- Implemented an automated evaluation pipeline utilizing LLM-as-a-judge against structured sales rubrics (objection handling, compliance, product knowledge) to deliver granular post-session scorecards and actionable coaching.

MULTI-MODAL SARCASM DETECTION & LLM FEATURE ABLATION | [Link](#)

2024

- Technologies: Python, PyTorch, ViSoBERT, CafeBERT, Hugging Face
- Researched and implemented multi-modal sarcasm detection on Vietnamese social media, achieving 2nd Place in the UIT Data Science Challenge 2024.
- Conducted extensive ablation studies on intermediate layer representations across ViSoBERT, CafeBERT, and GPT backbones, discovering that intermediate encoder representations with max pooling capture multi-layered figurative semantics significantly better than vanilla [CLS] token pooling.

SKILLS

AI & LLM Frameworks LangChain, LangGraph, Agentic Orchestration, RAG & Document Graphs, LLM-as-a-Judge, NLP, Multimodal AI (D-ID, ElevenLabs, Whisper), Pydantic, Vector Search, PyTorch, Hugging Face, Transformers

Programming Languages Python, Java, Rust, SQL, FastAPI, Spring Boot

Databases & Message Queues Milvus, Qdrant, PostgreSQL, MySQL, MongoDB, SQLite, Redis, Kafka, RabbitMQ

Infrastructure & DevOps Docker, Docker Compose, Azure, VPS, GitLab CI/CD, GitHub Actions

Languages Vietnamese (Native), English (Proficient (TOEIC 980 LR, 320 SW)), French (Independent (DELF B2))

CERTIFICATIONS

Nvidia Certificate of Competency — AI for Anomaly Detection, NVIDIA

2025

AWARDS

2024 2nd Place - UIT Data Science Challenge 2024, Won 2nd Place out of competitive teams by engineering a multi-modal sarcasm detection pipeline on Vietnamese social media; conducted in-depth feature ablations demonstrating that intermediate encoder representations significantly outperform standard [CLS] embeddings by preserving subtle ironic nuances lost in deeper arithmetic transformations.